

Practical Practice Sheet - 3

Object Oriented Programming - Simple Java Answers

Q1. String to Integer (myAtoi)

This program removes leading spaces, checks + or -, reads digits, stops at a non-digit, and keeps the answer inside the 32-bit integer range.

```
import java.util.*;

public class Main {
    static int myAtoi(String s) {
        int i = 0, sign = 1;
        long num = 0;

        while (i < s.length() && s.charAt(i) == ' ')
            i++;

        if (i < s.length() && (s.charAt(i) == '+' || s.charAt(i) == '-')) {
            if (s.charAt(i) == '-') sign = -1;
            i++;
        }

        while (i < s.length() && Character.isDigit(s.charAt(i))) {
            num = num * 10 + (s.charAt(i) - '0');

            if (sign == 1 && num > 2147483647)
                return 2147483647;
            if (sign == -1 && -num < -2147483648L)
                return -2147483648;

            i++;
        }

        return (int)(sign * num);
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int t = sc.nextInt();
        sc.nextLine();

        while (t-- > 0)
            System.out.println(myAtoi(sc.nextLine()));
    }
}
```

Sample output:

```
98
-456
45
0
-2147483648
```

Q2. Chess Championship - Carlsen Prize

C = Carlsen win (2 points), N = Chef win (0 points), D = draw (1 point each).
If Carlsen has more points, prize = 60*X. If tied, Carlsen gets 55*X.

```
import java.util.*;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int t = sc.nextInt();

        while (t-- > 0) {
            int x = sc.nextInt();
            String s = sc.next();

            int carlsen = 0, chef = 0;

            for (char c : s.toCharArray()) {
                if (c == 'C')
                    carlsen += 2;
                else if (c == 'N')

```

```
        chef += 2;
    else {
        carlsen++;
        chef++;
    }

    if (carlsen > chef)
        System.out.println(60 * x);
    else
        System.out.println(55 * x);
}
}
```

Sample output:

```
6000
24000
1650
40
```

Q3. Largest Odd Substring

The largest odd substring ends at the first odd digit found from the right.
So find the last odd digit and print everything from the beginning to that digit.

```
import java.util.*;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        String s = sc.next();

        int i = s.length() - 1;

        while (i >= 0) {
            int d = s.charAt(i) - '0';

            if (d % 2 == 1) {
                System.out.println(s.substring(0, i + 1));
                return;
            }

            i--;
        }

        System.out.println("-1");
    }
}
```

Example:
Input: 128764
Output: 1287

Q4. Convert String to Title Case

For each word, capitalize the first letter. If the whole word is already uppercase, keep it unchanged as an acronym.

```
import java.util.*;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int t = Integer.parseInt(sc.nextLine());

        while (t-- > 0) {
            String s = sc.nextLine();
            String[] words = s.split(" ");

            for (int i = 0; i < words.length; i++) {
                String w = words[i];

                if (!w.equals(w.toUpperCase())) {
                    w = w.substring(0, 1).toUpperCase()
                        + w.substring(1).toLowerCase();
                }

                words[i] = w;
            }

            System.out.println(String.join(" ", words));
        }
    }
}
```

Sample:
hello world -> Hello World
this is a CODECHEF problem -> This Is A CODECHEF Problem

Q5. Convert String to Number Without Parsing

Build the number digit by digit. For every digit, multiply the current number by 10 and add the new digit.

```
import java.util.*;

public class Main {
    static int stringToNumber(String s) {
        int num = 0;

        for (int i = 0; i < s.length(); i++) {
            int digit = s.charAt(i) - '0';
            num = num * 10 + digit;
        }

        return num;
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int t = sc.nextInt();

        while (t-- > 0) {
            String s = sc.next();
            System.out.println(stringToNumber(s));
        }
    }
}
```

Sample output:

```
123
42
1001
```

Note: These answers are simplified versions based on the questions in the uploaded Practical Practice Sheet - 3.